

Payment Profitability, Data Trust & Executive Decision

Why is payment contribution profit per successful transaction falling even though payment volume is increasing, and what should AstraPay change in the next 60 days?

The answer, in one line

Dilution, not deterioration. 88.6% of the fall is the mix of ticket sizes moving.

Three true statements that look like a contradiction

Every stakeholder can cite a number and be right. That is what makes this a diagnosis problem, not a reporting problem.



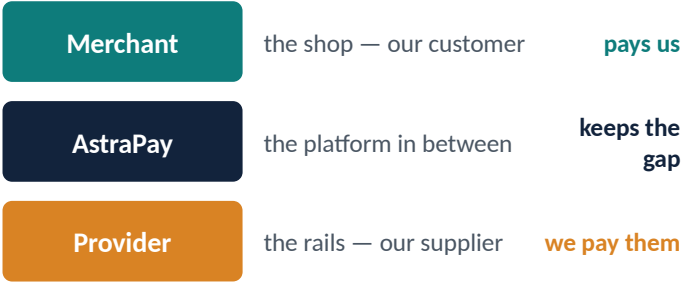
The resolution

Dilution, not deterioration	88.6% of the fall is ticket-size mix shifting. Almost every segment held its own economics or improved.
Growth landed in the wrong place	A \$14-median digital-goods cohort went 8.1% → 33.9% of successful transactions, while TRAVEL, ELECTRONICS and GROCERY lost share.
We routed and priced it as if it were large	Sub-\$90 traffic drifted onto the provider whose fixed fee is built for \$500 tickets — and that fee rose 50.6% mid-year.
Three of four stakeholder theories are wrong	Pricing, fraud and chargebacks were each tested and disproved. One was disproved by a data-quality artefact, not a business fact.

The business, the question, and the KPI it is asked in

AstraPay is a cross-border payments company in India, Singapore and the UAE. It earns a fee from merchants and pays a fee to providers.

Who pays whom



The executive KPI

contribution per successful txn

$$= \frac{(\text{merchant revenue} - \text{processing cost} - \text{risk cost})}{\text{successful transactions}}$$

USD at the SETTLEMENT rate

SMART, in short

- Size the drivers of the decline across 2025, separating business performance from data artefacts.
- Attribute at least 90% of the fall to named drivers in USD per txn, residual stated.
- Five-day diagnostic; actions startable and measurable inside 60 days.

KPI element	Defined to the row
Numerator	Merchant revenue – processing cost – risk cost, in USD
Denominator	Count of transactions with status CAPTURED
Date basis	transaction.created_at — the month we transacted, not the month cash settled
Exclusions	REVERSED out of both sides; negatives on CAPTURED with no parent quarantined; declined attempts leave the denominator

Four competing explanations, offered before anyone looked at the data

CFO

"Merchants negotiated us down on price."

H1b

Head of Payments

"Provider latency is destroying conversion."

H2 / volume

Risk Head

"The new fraud model declines good payments."

H3b

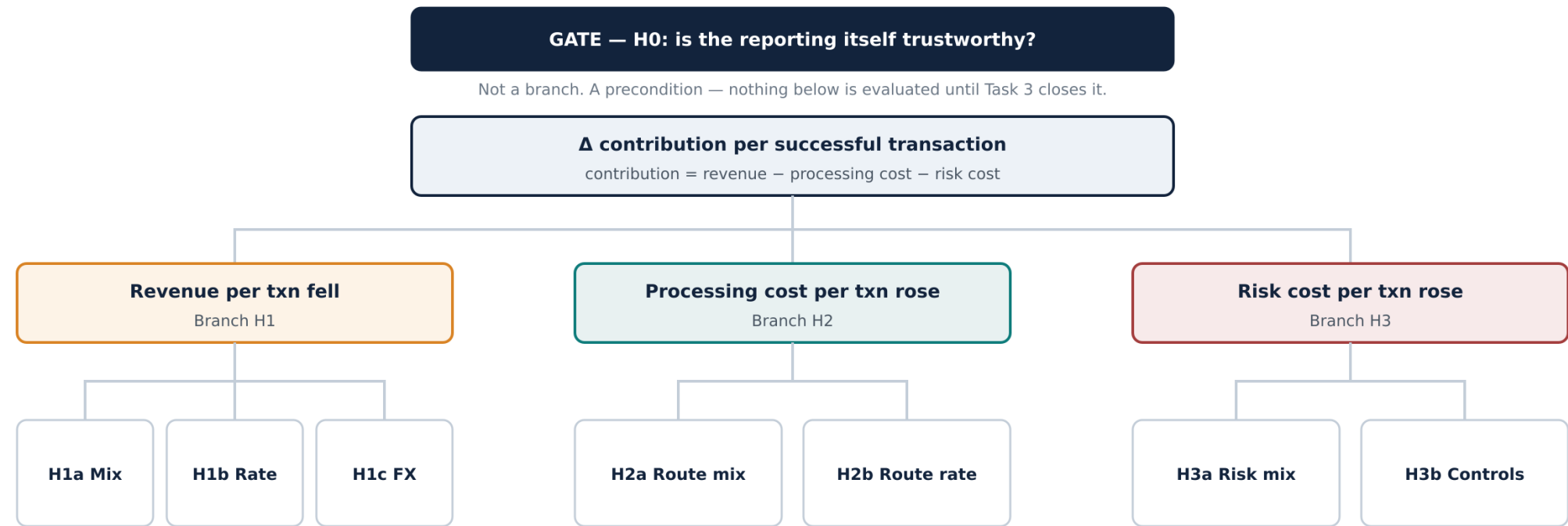
Data Head

"None of the reporting can be trusted anyway."

H0 — the gate

The hypothesis tree comes from the arithmetic, not from the room

Deriving the tree from the KPI formula is what makes it mutually exclusive and collectively exhaustive.



Why this is MECE

The tree is derived from the arithmetic of the KPI, not from stakeholder opinion. A fall in contribution must come from lower revenue, higher processing cost, higher risk cost, or a combination — there is no fourth option. Each branch then splits exactly two ways: the weights moved (mix), or the rate moved (within).
Every stakeholder claim maps onto a leaf. Nothing a stakeholder can say falls outside the tree, and no two leaves overlap.

Seven hypotheses, each with its falsification condition written first

Stating what would reject a hypothesis before running the test is what stops the analysis becoming a search for confirmation.

ID	Hypothesis	Confirms if	Rejects if
H0	The deterioration is partly a data artefact	Transaction and settlement totals fail to reconcile	Reconciliation closes and the defect log explains the gap
H1a	Revenue per txn fell because the book shifted	Segment weights shift materially across the period	Weights are stable, or reweighting removes the fall
H1b	Merchants were repriced downward	Take rate falls within a pricing plan	Take rate within every plan is flat or rising
H1c	Currency translation eroded revenue	Constant-FX contribution diverges from actual	The two series track closely
H2a	Route mix moved to a costlier provider	Route mix shifts and fixed-fee share rises	Route mix is stable
H2b	A provider raised its own unit cost	Holding route and ticket band fixed, cost per txn rises	Within route and band, cost per txn is flat
H3a	Risk mix moved to higher-risk bands	Risk-band weights shift upward	Risk-band weights are stable
H3b	Fraud controls decline good payments	At a comparable threshold, decline rates rise	Decline rates at a comparable threshold are flat

H3b carries a warning from the framing document: fraud scores come from more than one model version, so any comparison of score distributions across the cutover is invalid until the scales are normalised. That warning turned out to be the whole answer to Task 6.

What is out of scope, and the six assumptions that could be argued with

Each assumption is a decision with a defensible alternative. Naming them lets a reviewer disagree precisely instead of vaguely.

Out of scope

- Fixed and overhead cost — contribution margin, not a full P&L.
- Acquisition and merchant onboarding cost.
- Settlement records with no matching transaction — reported as a reconciliation break, not analysed.
- Any period outside 2025.

Stated up front so an absent finding is not read as a null result

- Causation between latency and merchant churn — no retention data exists.
- Whether a declined payment was lost revenue or a customer who retried — retry linkage is not in the data.
- Merchant profitability net of relationship and support cost.
- Price elasticity — no historical price change exists to estimate it from.

#	Assumption	Why, and the alternative that was rejected
A1	FX uses the SETTLEMENT rate on the transaction date	MID is quoted, not realised. The spread widens 0.35% → 0.60% in Q4, so the error grows over exactly the period under investigation.
A2	Settlement fees attributed to the transaction month	Keeps revenue and cost in one period. Settlement-month attribution would smear the August incident across two months.
A3	Captured transactions with no settlement row are retained	4.77% of rows, weighted to December. Dropping them deletes the newest month's revenue and exaggerates the decline.
A4	Missing merchant_id rows stay in the executive KPI	They are real payments. Excluded only from merchant-level breakdowns, where they would bias the answer.
A5	Negative amounts with status REVERSED are legitimate	A reversal is a business event. Only negatives on CAPTURED with no parent are impossible.
A6	Route costs are read as-of the transaction date	PROV_ORBIT reprices on 1 July. A single latest-rate join would erase the step change entirely.

Eleven files, and the grain of each one

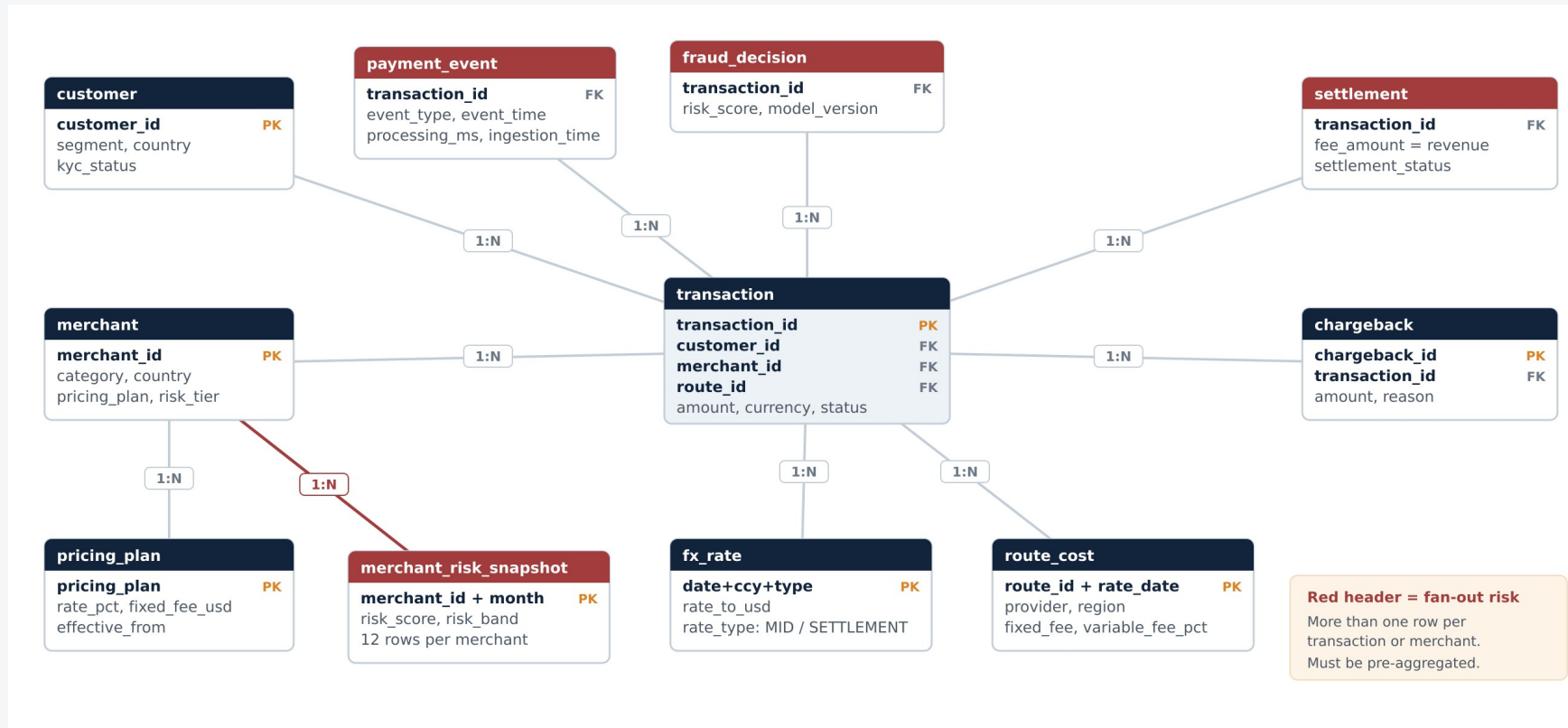
Grain is the first question, not a detail. Six of these carry more than one row per transaction, and every one of those is a way to inflate the answer.

File	Rows	Grain	What it carries	Join hazard
transaction	260,287	1 per payment attempt	amount, currency, channel, status, route_id	— the spine
payment_event	~1.00M	2–4 per transaction	lifecycle events, processing_ms, ingestion_time	FAN-OUT
fraud_decision	466k	1–4 per transaction	risk_score, decision, model_version	FAN-OUT ×1.8
settlement	224k	1–2 per transaction	fee_amount = AstraPay's revenue	FAN-OUT
chargeback	721	1 per dispute	disputed amount, reason, opened_at	one-to-many
merchant_risk_snapshot	7,200	12 per merchant	risk_score, risk_band by month	FAN-OUT ×11.9
route_cost	dated	1 per route per rate_date	fixed_fee, variable_fee_pct	dated — as-of join
fx_rate	dated	1 per date × currency × rate_type	rate_to_usd, MID or SETTLEMENT	two rate types
merchant	600	1 per merchant	category, country, pricing_plan, risk_tier	safe
customer	15,000	1 per customer	segment, country, kyc_status	safe
pricing_plan	4	1 per plan	rate_pct, fixed_fee_usd, billing_basis	safe

settlement.fee_amount is the fee AstraPay charges the merchant — revenue. The fee AstraPay pays the provider is a different number and lives in route_cost. Reading fee_amount as cost makes contribution zero by construction.

How the eleven files relate, and the three controls that keep them honest

Red headers carry more than one row per key. Every one of them is pre-aggregated before it touches the fact table.



As-of join

merchant_risk_snapshot → latest snapshot at or before the transaction month. Naive: ×11.92 GPV inflation.

Pre-aggregate first

fraud_decision, payment_event, settlement, chargeback all collapsed to one row per transaction_id before joining. Naive fraud join: ×1.79.

Dated join, explicit rate type

route_cost and fx_rate joined on the transaction date, with rate_type = SETTLEMENT. Key-only joins multiply by every rate date.

Thirteen defects, and why "drop the bad rows" is the wrong instinct

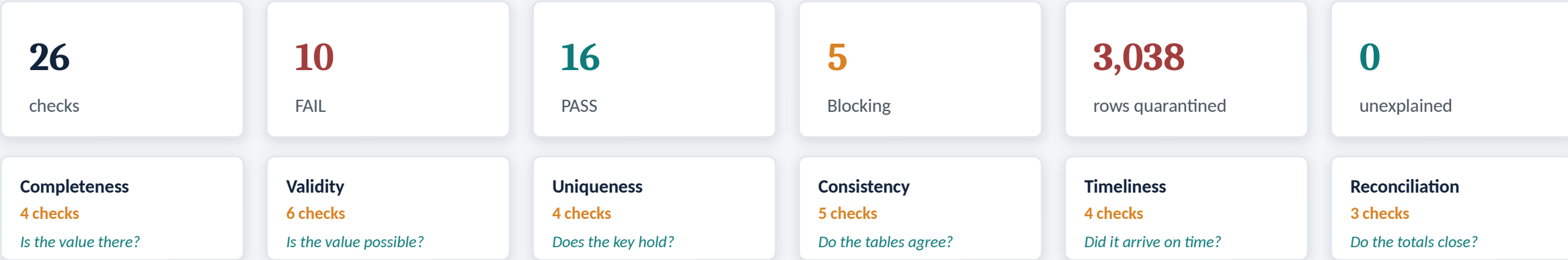
Three remediation verbs, applied deliberately: quarantine, repair, or retain-with-flag. Deletion is the option that silently changes the answer.

#	Defect	Measured	Severity	Remediation
1	Missing merchant_id	0.68% overall — but WALLET 2.10% vs 0.41% elsewhere	WARNING	Retain with flag. Not uniform, so dropping biases channel analysis
2a	Reversals: negative amount, status REVERSED	1,952 rows	INFO	Legitimate business event — net them, do not drop
2b	Orphan negatives on CAPTURED, no parent	390 rows	BLOCKING	Quarantine — an impossible state
3	risk_score > 1	37.9% of fraud rows, all model_version v3.0	WARNING	Normalise by version before any threshold comparison
4	Latency deterioration	PROV_MERIDIAN, 1 Aug – 24 Sep	WARNING	Flag as an incident, exclude from trend fitting
5	Late-arriving events	4.44% ingested >2 days late, max 9 days	WARNING	event_time drives KPIs; ingestion_time measures reporting lag
6a	Captured transactions with no settlement row	4.77%, weighted to December	BLOCKING	Retain with flag; impute from the rate card, never treat as zero
6b	Transactions with two settlement rows	1,334 (partial + CORRECTION)	BLOCKING	Sum to transaction grain before joining
6c	Settlement rows with no matching transaction	447 rows	BLOCKING	Exclude and report to Finance Ops as a reconciliation break
7	Two FX rate types	MID and SETTLEMENT for every date	WARNING	Pick one, state why, sensitivity-test the other
8	merchant_risk_snapshot fan-out	12 rows per merchant — ×11.9 inflation	BLOCKING	As-of join on snapshot_month
9	fraud_decision fan-out	1–4 rows per txn — ×1.8 inflation	BLOCKING	Pre-aggregate to transaction grain
9b	payment_event fan-out	2–4 rows per transaction	BLOCKING	Pivot before joining

Note the asymmetry in defect 1: the missing merchant rate is five times higher on WALLET than anywhere else. A defect that is not uniformly distributed cannot be dropped without biasing whatever dimension it correlates with.

Twenty-six checks across six dimensions: 10 failed, 16 passed

The gate from Task 1: until H0 is closed, no other hypothesis is worth testing. This is how it was closed.



Dimension	Check that failed	Failed	Severity	Consequence if ignored
Uniqueness	more than one settlement row per transaction	1,334	Blocking	Merchant revenue double-counts on those transactions
Uniqueness	merchant_risk_snapshot composite key not unique	12 / merchant	Blocking	×11.92 inflation on every merchant-level measure
Validity	risk_score outside 0 to 1	37.9%	Blocking	The fraud hypothesis looks true when it is not
Validity	negative amount with status CAPTURED	390	Blocking	Contribution understated by an impossible state
Consistency	settlement.transaction_id not found in transaction	447	Blocking	Revenue attributed to payments that do not exist
Completeness	transaction.merchant_id is null	0.68%	Warning	Channel analysis biases if dropped — WALLET is 2.10%
Reconciliation	captured transaction with no settlement row	4.77%	Warning	December revenue disappears if treated as zero

Verdict on H0: the reporting is not trustworthy as delivered, and it is repairable. The Data Head is right about the symptom and wrong about the conclusion — the gate opens once these ten are remediated, it does not stay shut.

Run the wrong join on purpose, then show the right one

A control that is asserted can be wrong. A control that is demonstrated numerically cannot be waved away in a review.

```
# deliberately naive – the join a hurried analyst writes
naive = tx.merge(risk_snap, on="merchant_id", how="left")
print(naive.amount_usd.sum() / tx.amount_usd.sum())
# -> 11.92      twelve snapshots per merchant

# correct: as-of join, latest snapshot at or before the month
snap = (risk_snap[risk_snap.snapshot_month <= tx_month]
        .sort_values("snapshot_month")
        .groupby("merchant_id").tail(1))
# -> 1.00      grain preserved
```

×11.92

GPV inflation, risk snapshot

×1.79

GPV inflation, fraud join

Why this matters more than it looks

An inflated total is obvious. An inflated total that is inflated EVENLY is not — ratios still look sane, trends still point the same way, and the error survives review. ×11.92 on gross value with the transaction count untouched is exactly that shape.

Quarantine reconciliation — nothing disappears silently

3,038

rows excluded from the KPI

=

2,668

duplicate settlement rows

+

370

impossible negative amounts

+

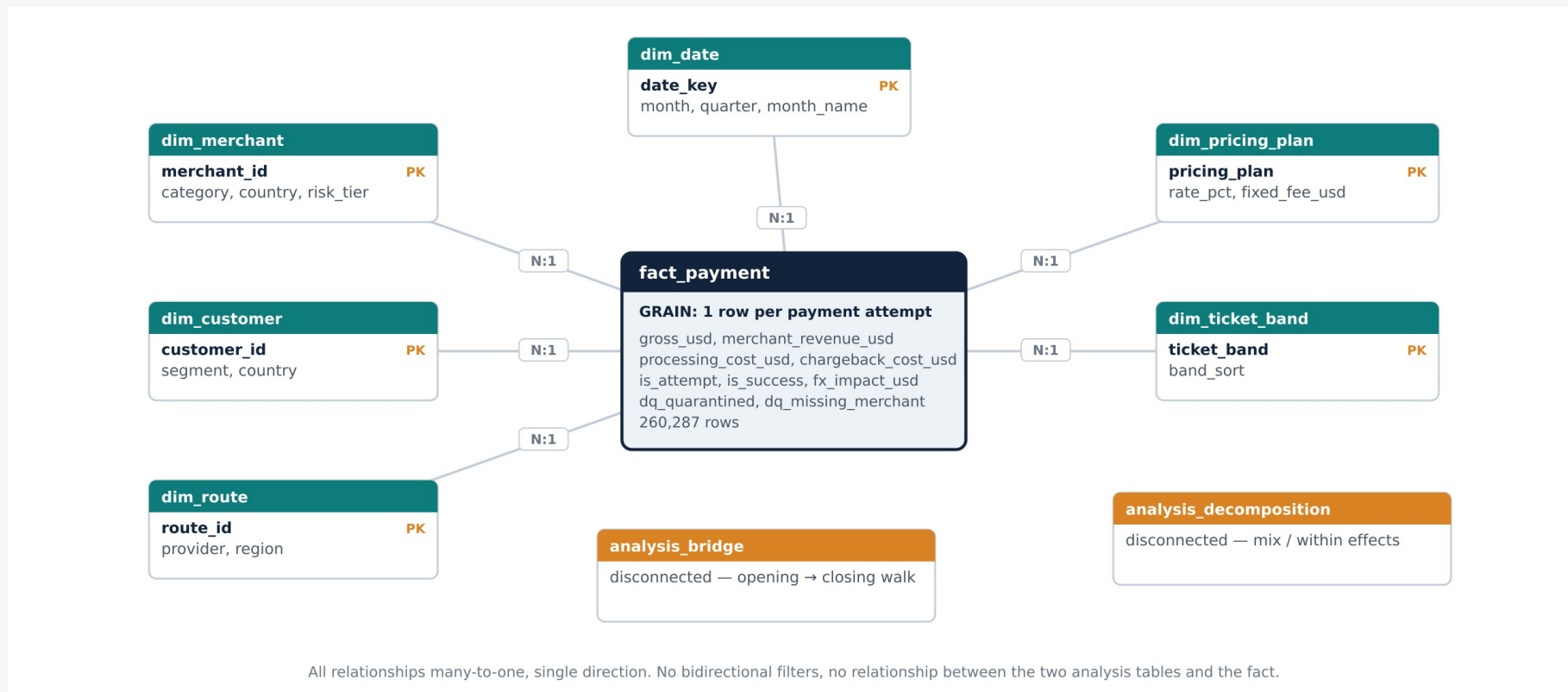
0

unexplained

```
# every check emits the same seven fields, so the scorecard is a table, not a paragraph
add("Uniqueness", "more than one settlement row per transaction", "settlement",
    (s.groupby("transaction_id").size() > 1).sum(), s.transaction_id.unique(),
    "Blocking", "Sum fee_amount to transaction grain before joining") # + 25 more
```

One fact, six conformed dimensions, two disconnected analysis tables

Everything downstream — the notebooks, the dashboard, this deck — reads from this model and nothing reads from the raw files.



The rule the whole pipeline is built around

Every one-to-many source is collapsed to one row per transaction BEFORE it is joined to the spine — not after, not with a DISTINCT, not with a window function at query time. Each staging function carries a GRAIN docstring, and that docstring is the contract: if a function claims one row per transaction_id, a single assertion proves it. This is the difference between a model that is correct and a model that happens to give the right answer on this month's data.

Staging, the contribution formula, and the row type that nearly broke it

t4_core.py — four staging functions collapse the hazardous sources; the fact build encodes every Task 1 decision as a line of code.

```
def stg_settlement(s):
    """GRAIN: 1 row per transaction_id.
    Collapses partial + correction pairs."""
    return (s.groupby("transaction_id")
            .agg(merchant_fee_usd=("fee_amount", "sum")))

def stg_fraud(f): # GRAIN: 1 row per transaction_id
    f["score_norm"] = np.where(
        f.model_version.str.startswith("v3"),
        f.risk_score / 100.0, f.risk_score) # two scales

def stg_event(e) # pivots lifecycle, keeps latency
def stg_chargeback(c) # sums disputes to txn grain
```

```
fx_sel = fx[fx.rate_type == RATE_TYPE] \
    .set_index(["currency", "rate_date"]).rate_to_usd
t["gross_usd"] = t.amount * t.fx_rate_to_usd

t["contribution_usd"] = ( t.merchant_revenue_usd
                        - t.processing_cost_usd
                        - t.chargeback_cost_usd )

# no merchant -> no plan -> no rate -> NaN contribution
portfolio_rate = (t.loc[settled, "merchant_fee_usd"].sum()
                  / t.loc[settled, "gross_usd"].sum())
imputed_rev = np.where(np.isnan(imputed_rev),
                       t.gross_usd * portfolio_rate, imputed_rev) # flagged
```

The decisions encoded above

Sum, never first

A partial settlement plus a CORRECTION row are two halves of one fee. First loses the correction, max loses the partial — only the sum is the fee.

Normalise at the boundary

If the score scale is fixed downstream, someone eventually queries the raw column and gets a different answer. Fix it once and never expose the raw value.

Dated joins, explicit rate type

SETTLEMENT FX on the transaction date; route cost as-of the transaction date. PROV_ORBIT steps \$0.35 → \$0.52 on 1 July — a key-only join averages that away.

Quarantine means flagged, not deleted

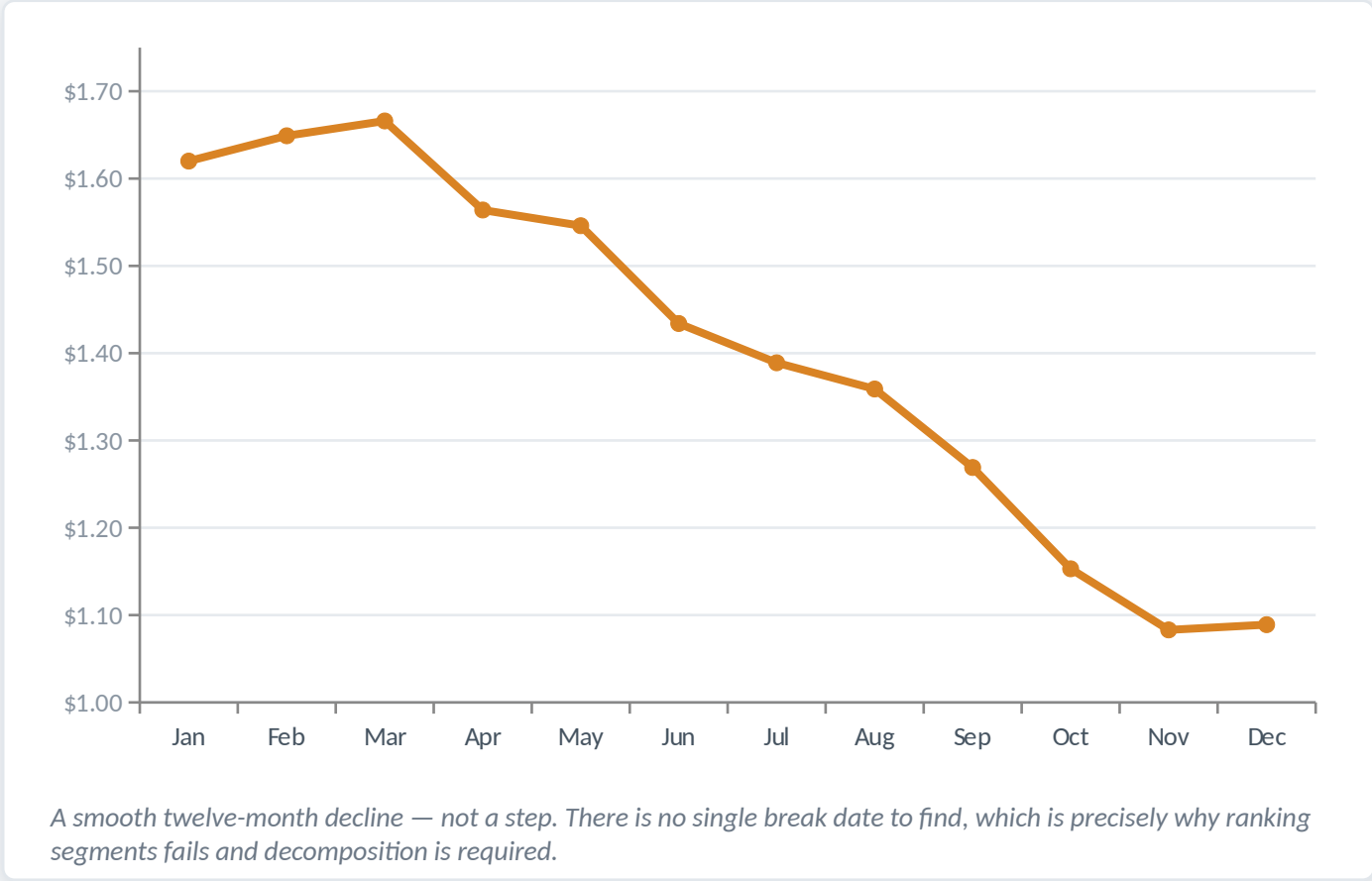
The 390 impossible rows stay in the fact table with dq_quarantined = TRUE. Measures filter them; the count stays reportable.

The NaN that broke the decomposition

Rows with no merchant had no imputable revenue, so contribution was NaN and the residual was non-zero. The portfolio-rate fallback — flagged, never silent — took it to exactly 0.

Validation, and the monthly series the whole case rests on

validation_checks() is the proof that the model does not double count. It runs every time the notebook runs.



Validation checks that must pass before any analysis

Fact row count = transaction row count 260,287 = 260,287	PASS
transaction_id unique in the fact 0 duplicates	PASS
Gross USD equals the source total ratio 1.000	PASS
Every staged table is at txn grain 4 of 4 assertions hold	PASS
Revenue – cost – risk = contribution residual 0.00	PASS

Jan \$1.620 – Dec \$1.089 –32.8%

Ranking segments cannot answer this question. Decomposition can.

Two questions a ranked table never answers: how much of the fall does a segment contribute, and is that its weight moving or its own economics moving?

The identity

$$\Delta c = \sum (w_1 - w_0) \cdot c_0 \quad \leftarrow \text{MIX}$$

$$+ \sum w_1 \cdot (c_1 - c_0) \quad \leftarrow \text{WITHIN}$$

w = segment share of successful txns
c = contribution per txn in that segment

the two terms sum to Δc with NO residual

MIX effect

The segment's weight changed. Its own economics may be untouched — the portfolio average moves because the composition moved.

Commercial question: what are we selling, and to whom?

WITHIN effect

The segment's own contribution per transaction changed. Same customers, same products, worse or better unit economics.

Operational question: what changed in how we run it?

```
def _effects(f, dim, first, last):
    a, b = f[f.txn_month == first], f[f.txn_month == last]
    idx = sorted(set(a[dim]) | set(b[dim])) # union!
    w0 = a[dim].value_counts(normalize=True).reindex(idx).fillna(0)
    w1 = b[dim].value_counts(normalize=True).reindex(idx).fillna(0)
    c0 = a.groupby(dim).contribution_usd.mean().reindex(idx).fillna(0)
    c1 = b.groupby(dim).contribution_usd.mean().reindex(idx).fillna(0)
    return (w1 - w0) * c0, w1 * (c1 - c0) # mix, within
```

Union of both indexes

A segment present in December but absent in January must appear with weight 0, not be dropped — otherwise the weights stop summing to 1 and the residual stops being zero.

Rounding is display-only

Rounding inside the calculation is how a residual of 0 becomes 0.0003 and a reviewer stops trusting the identity.

Missing merchants get their own segment

"(unknown)" is a segment, not a dropped row. Dropping it breaks the identity and quietly deletes 0.68% of real payments.

If the fall were a within effect the plan would be operational: fix the process, retune the model, escalate to the provider. If it is mix, none of that helps — every segment could be performing perfectly and the average would still fall.

Nine dimensions decomposed. One of them answers the question.

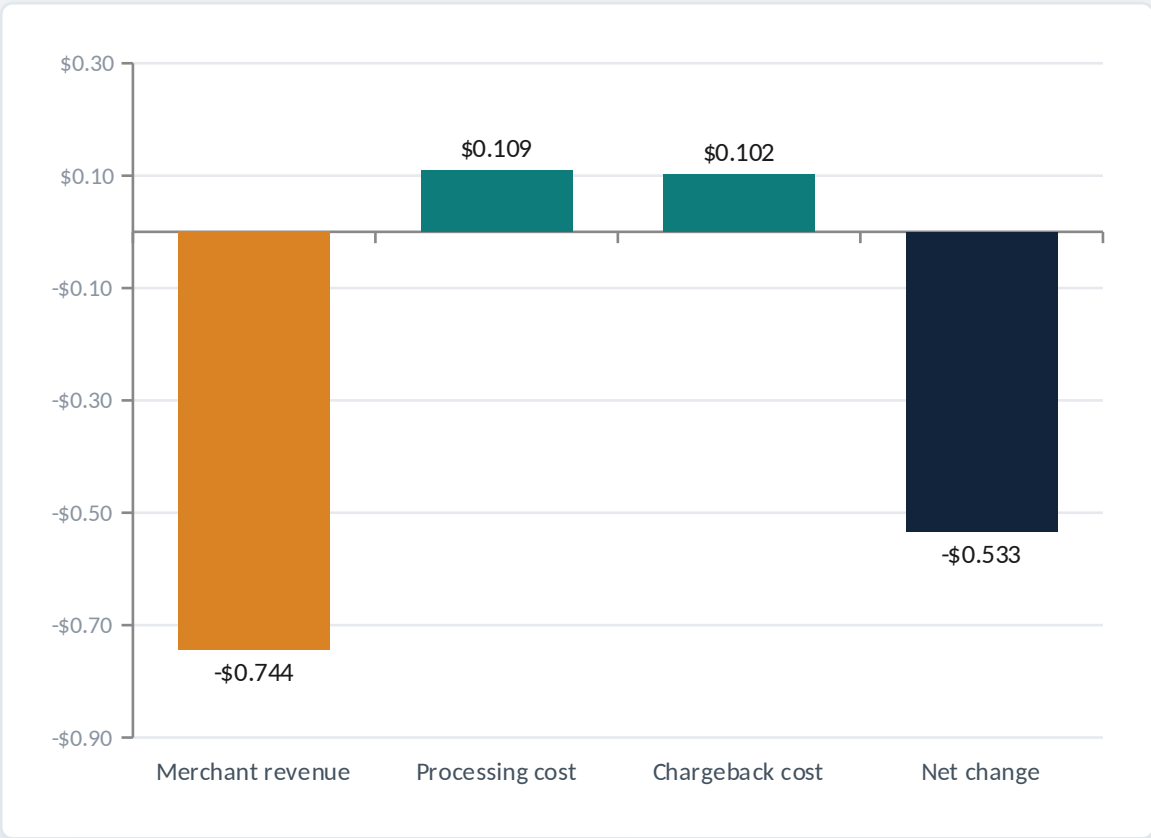
Every row sums to the same actual change of −\$0.5310, with residual zero. That is the check that the arithmetic is right before the interpretation starts.

Dimension	Mix effect	Within effect	Sum	Mix share	Residual	Reading
Ticket size band	−\$0.4719	−\$0.0609	−\$0.5310	88.6%	0.000000	The answer. Small tickets took over the book.
Category × country cohort	−\$0.4296	−\$0.1014	−\$0.5310	80.9%	0.000000	Same story named differently — DIGITAL_GOODS in SG and AE
Merchant category	−\$0.41	−\$0.12	−\$0.5310	~77%	0.000000	Consistent; the cohort view is sharper
Provider	small	moderate	−\$0.5310	mixed	0.000000	Routing drift shows up here as a within effect
Payment channel	moderate	small	−\$0.5310	mostly mix	0.000000	WALLET grows with the small-ticket cohort
Merchant country	moderate	small	−\$0.5310	mostly mix	0.000000	SG and AE gain share
Customer segment	small	small	−\$0.5310	—	0.000000	Not where the story lives
Merchant risk band	small	small	−\$0.5310	—	0.000000	Disproves H3a — risk weights are stable
Pricing plan	moderate	≈ 0	−\$0.5310	~all mix	0.000000	Disproves H1b — no plan's own rate fell

Ticket band and the category-country cohort are two views of one phenomenon, which is why they agree: the cohort that grew is the cohort with a \$14 median ticket. Reporting both is the honest thing to do — reporting only the higher number would be shopping for a headline.

What grew, what shrank, and which part of the formula actually moved

Decomposing the same fall a second way — by formula component rather than by segment — and getting a consistent answer is the cross-check.



Cohort	Weight Jan	Weight Dec	Contrib/txn Jan	Contrib/txn Dec
DIGITAL SME, SG+AE	8.1%	33.9%	\$0.083	-\$0.138
TRAVEL	12.4%	9.0%	\$5.06	\$4.72
ELECTRONICS	7.1%	5.2%	\$4.09	\$4.66
GROCERY	30.2%	21.6%	\$1.20	\$1.03
SERVICES	18.9%	13.2%	\$1.16	\$1.23
GAMING	15.0%	11.2%	\$0.44	\$0.37
DIGITAL_GOODS (other)	8.3%	5.9%	\$0.57	\$0.71

Revenue did all the damage

-\$0.744 per transaction. A percentage fee returns about \$0.48 on a \$14 ticket and \$12 on a \$600 one.

Costs actually improved

Processing cost per txn fell \$1.08 → \$0.97, chargeback cost \$0.25 → \$0.15. Both are positive bars, so the cost story is dead.

Three of seven cohorts improved

Exactly one goes negative — and it is the one that quadrupled its share. That is what a mix problem looks like from the inside.

FX is the memo line: drag widens -\$0.01 → -\$0.09, about 16% of the fall. Real, secondary, and already inside the revenue bar — not added on top of it.

Six hypotheses tested. Three of the four in the room are wrong.

Every test holds composition constant — without that, a segment gaining share looks exactly like a segment getting worse.

Driver	Type	Claimed by	Test result	Effect / txn	Verdict
Merchant pricing	Commercial	CFO	No pricing plan's take rate fell. FLAT_2_0 rose 0.2247pp. Blended rate falls only because plan mix moved.	\$0.000	DISPROVED
Fraud controls	Risk	Risk Head	On the normalised score, decline rate moves -0.014pp. The apparent spike is a score-scale artefact at the v2.1 → v3.0 cutover.	≈ \$0.000	DISPROVED
Chargebacks	Risk	Risk Head	Chargeback cost per successful txn falls \$0.2515 → \$0.1495. Rate ranks high, impact ranks last.	+\$0.102	DISPROVED
Provider latency	Operational	Head of Payments	PROV_MERIDIAN p95 227 → 1,517ms, success 90.3% → 82.5% for eight weeks. Roughly 698 transactions lost.	≈ \$0.000	REAL, WRONG KPI
FX translation	External	—	Drag widens -\$0.01 → -\$0.09 as the settlement spread goes 0.35% → 0.60% in Q4.	-\$0.08	SECONDARY
Routing drift + Orbit repricing	Operational	Our finding	Cohort share on PROV_ORBIT climbs 28% → 89% across the year while Orbit's fixed fee rises 50.6% on 1 July.	-\$0.10	CONFIRMED
Ticket-size mix	Commercial	Our finding	Mix -\$0.4719 against within -\$0.0609 on ticket band; -\$0.4296 / -\$0.1014 on the cohort view.	-\$0.43	CONFIRMED

Sizing beats ranking. Chargebacks rank near the top on rate and last on impact; latency is the most dramatic operational event of the year and contributes almost nothing to this KPI. A ranked list would have put both at the top of the plan.

The fraud hypothesis is disproved by a data defect, not a business fact

The most important finding in Task 6 — and it is a data-quality finding that happens to answer a commercial question.

model_version	Share of fraud rows	min score	max score	Scale	What a naive reading says
v2.1	62.1%	0.00	1.00	0 – 1	Baseline
v3.0 (from 1 Sep)	37.9%	0.00	100.00	0 – 100	"Risk scores exploded in September"

The trap

Compare score distributions across 1 September and the mean risk score rises roughly a hundredfold. Compare decline rates at a fixed threshold of 0.8 and everything above 0.8 after the cutover is a v3.0 score — which is almost all of them. The evidence for "the new model is rejecting good payments" is overwhelming, internally consistent, and entirely an artefact of two scales sharing one column.

After normalising

Divide v3.0 scores by 100 and the distributions line up. Decline rate at a comparable threshold moves -0.014pp across the cutover — declines fall very slightly. Chargeback rate does not rise, which it would if controls had loosened, and does not fall, which it would if they had tightened. The model is behaving.

Why this belongs in an executive deck and not a data-quality appendix

Without Task 3, this hypothesis is confirmed with a chart any reviewer would accept, and the 60-day plan contains a fraud-model rollback that costs engineering time and raises genuine fraud losses. The check that prevents it — "is risk_score inside its declared range?" — takes one line to write and is exactly the kind of check that gets skipped.

One real incident, correctly parked. One real driver, correctly promoted.

The discipline being tested here is sizing an effect rather than reacting to its severity.

PROV_MERIDIAN latency incident — 1 Aug to 24 Sep

	Jul	Aug	Sep	Oct
p95 processing_ms	227	1,517	1,417	227
Success rate	90.3%	82.5%	84.2%	89.9%

Severe, genuine, worth roughly 698 lost transactions. It costs VOLUME. It barely touches contribution per SUCCESSFUL transaction, because a failed attempt never enters the denominator and incurs almost no cost. If a decomposition assigns this a slice of the ~\$0.53, the model is wrong.

Routing drift + the 1 July Orbit repricing

Small-ticket cohort	Q1	Q2	Q3	Q4
Share routed to ORBIT	28%	47%	67%	89%

The routing table keys on merchant COUNTRY, not ticket size. As the small-ticket cohort grows in SG and AE, more micro-payments land on the provider with the highest fixed fee — and that fee rose from \$0.35 to \$0.52 on 1 July, visible as a dated step in route_cost.csv.

The break-even that turns this into a routing rule

Provider	Fixed fee	Variable	Cost on a \$14 ticket	Cost on a \$600 ticket	Where it wins
PROV_NORTH	\$0.05	0.60%	\$0.13	\$3.65	Everything below ~\$147
PROV_MERIDIAN	\$0.15	0.45%	\$0.21	\$2.85	Middle of the range
PROV_ORBIT	\$0.52 (was \$0.35)	0.28%	\$0.56	\$2.20	Large tickets only

Break-even between NORTH and ORBIT: about \$94 before the repricing, about \$147 after. The cohort that grew has a \$14 median ticket and was being routed to the provider that only wins above \$147.

Power BI consumes the governed star, never the raw files

Ten tables from powerbi_model/. Six many-to-one single-direction relationships. Two analysis tables deliberately disconnected.

Measure	DAX	Additive?
Attempted / Successful Volume	SUM(fact_payment[is_attempt]) / SUM(fact_payment[is_success])	Yes
Gross Payment Value	SUM(fact_payment[gross_usd])	Yes
Merchant Revenue	SUM(fact_payment[merchant_revenue_usd])	Yes
Processing / Chargeback Cost	SUM(fact_payment[processing_cost_usd]) / SUM(...[chargeback_cost_usd])	Yes
Contribution Profit	[Merchant Revenue] - [Processing Cost] - [Chargeback Cost]	Yes
Success Rate	DIVIDE([Successful Volume], [Attempted Volume])	NO
Contribution per Successful Txn	DIVIDE([Contribution Profit], [Successful Volume])	NO
Cost per Successful Txn	DIVIDE([Processing Cost], [Successful Volume])	NO
Quarantined Rows	CALCULATE(COUNTROWS(fact_payment), fact_payment[dq_quarantined] = TRUE())	Yes

The non-additive rule

- Every ratio is a DIVIDE of two additive measures, so it recomputes at any filter level instead of aggregating a pre-computed value.
- No ratio goes in a stacked visual, where a reader reads the parts as summing to the whole.
- Anything containing "per" or "Rate" is non-additive — the naming carries the signal.

Five build errors worth writing down

- Columns as Column1..Column5 — header row not promoted.
- DataFormat.Error on "True" — boolean flags arrive as text; convert Text → True/False → Whole Number.
- Circular dependency ticket_band ↔ ticket_band_sort — the sort column read the band it sorts.
- Blank KPI card — created as a calculated column, not a measure ("Sum of..." gave it away).
- Processing Cost \$91K instead of \$234K — a visual-level filter left over from an earlier experiment.

Page 1 — Executive overview

Five KPI cards plus the scissors chart

[PASTE POWER BI SCREENSHOT HERE]

Page 1 — Executive overview — full page capture

How to read this page

1

Attempted Volume, Successful Volume, Success Rate, Contribution Profit, and Contribution per Successful Txn — the last sized to dominate, because it is the KPI the executive question is asked in.

2

Visual 1: line and clustered column. Columns are Contribution Profit (rising). The line on the secondary axis is Contribution per Successful Txn (falling). The two crossing is the paradox in one picture.

3

Slicers for quarter, merchant category and provider, synced across all three pages.

Page 2 — Diagnostic

Where the decline comes from, and whether it is mix or performance

[PASTE POWER BI SCREENSHOT HERE]

Page 2 — Diagnostic — full page capture

Placeholder — replace the dashed frame with the Power BI screenshot of this page before submitting.

How to read this page

1

Visual 2: stacked bar from analysis_decomposition filtered to "Ticket size band", showing mix_effect against within_effect per band. The bars are almost entirely mix — that is the finding.

2

Visual 3: matrix of Cost per Successful Txn, provider on rows and ticket band on columns, with Successful Volume as a second value so the reader sees where volume sits. ORBIT \$0.53 under \$25 against NORTH \$0.14 is visible in one cell.

3

Supporting: ticket-band share over time, cohort economics crossing zero, and effective take rate by pricing plan — the last disproves the CFO at a glance.

Page 3 — Action

The bridge and the 60-day plan

[PASTE POWER BI SCREENSHOT HERE]

Page 3 — Action — full page capture

Placeholder — replace the dashed frame with the Power BI screenshot of this page before submitting.

How to read this page

- 1

Visual 4: waterfall from analysis_bridge. Opening \$1.613, merchant revenue −\$0.744, processing cost +\$0.109, chargeback cost +\$0.102. No closing row is supplied — Power BI computes its own total, so a hand-written closing row is counted twice.
- 2

The Task 6 verdict table beside it as a supporting table: driver, type, claimed by, result, verdict. It is the audit trail for the three disproved hypotheses.
- 3

Below: the five-action 60-day table imported from actions_60day.csv, so deck, dashboard and repository all quote the same five actions.

Price the floor, route the floor, defend the top

Small tickets are not the problem. Small tickets priced and routed like large ones are.

PRICE

\$0.35 minimum fee on FLAT_2_0

2.00% + \$0.20 returns ~\$0.48 on a \$14 ticket against \$0.56 of Orbit cost — loss-making by design.

ROUTE

Sub-\$90 volume to PROV_NORTH

Break-even against ORBIT is ~\$147 after the July repricing. Below it, NORTH costs \$0.13 where ORBIT costs \$0.56.

DEFEND

Over-\$500 volume, and renegotiate Orbit

TRAVEL and ELECTRONICS improved their own economics while losing share. Orbit's fixed fee rose 50.6% with no repricing on our side.

#	Action	Owner	Risk	Success metric	Window
1	Route transactions under \$90 to PROV_NORTH	Head of Payments	Capacity limits at North; needs a volume commitment check first	Cost per successful txn by ticket band	Days 1-20
2	Minimum per-transaction fee of \$0.35 on FLAT_2_0	CFO	Merchant churn among DIGITAL_GOODS SME accounts	Contribution per txn, DIGITAL_GOODS SG and AE	Days 1-30
3	Renegotiate the PROV_ORBIT fixed fee	Procurement	Orbit may refuse; contract may have no reopener until renewal	Fixed fee recorded in route_cost	Days 10-60
4	Protect large-ticket volume in TRAVEL and ELECTRONICS	Commercial	Slower headline growth; incentives reward count, not value	Share of successful volume above \$500	Days 15-60
5	Fix the wallet integration dropping merchant_id	Data Head	Engineering capacity; competing roadmap items	Missing merchant_id rate by channel	Days 20-45

Day-60 test: contribution per successful transaction stops falling and recovers toward \$1.30, with the under-\$25 cohort back above zero. If it has not moved, actions 1 and 2 were not the binding constraint and the pricing-plan analysis is re-opened. Not recommended: a fraud-model rollback, a chargeback programme, or a pricing renegotiation — all three were tested and none moves this KPI.

CONCLUSION

Dilution of a high-value book, not deterioration of it

Situation

Volume grew 70% across 2025, from 16,400 attempts in January to 27,903 in December, and total contribution profit grew 14%. On those two numbers the year was a success.

Complication

Contribution per successful transaction fell 33%, \$1.620 → \$1.089. Every additional transaction is worth a third less — and four stakeholders each had a different explanation, three of which were wrong.

Question

Is this a performance problem or a composition problem, and what should change in the next 60 days?

Answer

Composition. 88.6% of the decline is ticket-size mix shifting, not any segment performing worse — three of seven cohorts improved. A \$14-median digital-goods cohort went 8.1% → 33.9% of the book while TRAVEL, ELECTRONICS and GROCERY lost share. We then routed it down a pipe priced for large tickets and let its fixed fee rise 50.6% mid-year. The fix is a fee floor, a routing rule keyed on ticket size rather than country, and a defence of large-ticket volume.

Volume growth is not the same thing as value growth. This year AstraPay bought a lot of the first and sold some of the second.

What could make this wrong

Elasticity is unmeasured

No historical price change exists to estimate churn from. Launch the fee floor on the top 50 merchants and track churn weekly first.

Capacity at PROV_NORTH

The routing shift assumes NORTH absorbs ~34% more volume at the quoted rate. Confirm a volume commitment and re-quote before moving traffic.

A blind spot in the data

2.1% of wallet transactions carry no merchant, so cohort figures are computed on the remainder. Action 5 restores attribution; the decomposition is re-run when it lands.

Causality is inferred, not tested

The decomposition is exact arithmetic on what happened, not an experiment. Actions 1 and 2 launch on a subset so the effect is measured, not assumed.